# Sahaj Bhadja

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## Summary

MSCS candidate at USC with 2+ years building production **Python** services on **Docker**, **Kubernetes**, and **CI/CD** pipelines across **AWS** and **Azure**, with strong focus on databases, observability, and reliable infrastructure delivery.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Operating Systems, Database Systems, Analysis of Algorithms, Web Technologies

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, OOP, Operating Systems, Data Retrieval

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*

- Built production **Python** platform services on **FastAPI** and **PostgreSQL** for 500+ users with **CI/CD** releases, structured logging, and automated validation while improving latency from **1.2s to 350ms** and maintaining 99%+ reliability.
- Implemented ETL-style pipelines processing 5K+ monthly jobs with reconciliation checks and cloud-backed storage, accelerating incident triage through monitoring hooks and documented deployment workflows across production environments.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*

- Shipped production **Flask APIs** on **AWS** with **GitHub Actions CI/CD** for 150+ users and 10K+ daily requests, owning deployment validation, documentation, and stable performance during rapid iteration cycles.
- Participated in code reviews and pairing sessions, reducing feedback turnaround **60%** while independently debugging production issues and improving maintainability for cross-functional engineering partners.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*

- Engineered cross-platform software with **Jenkins CI/CD** and 20+ **PyTest**/Mockito suites across **Linux**, Windows, and macOS, boosting pipeline efficiency **68%** while supporting stable production delivery in Agile sprints.
- Troubleshot **AWS Glue** and S3 ETL integrations, improving response times **30%** and documenting deployment workflows to reduce manual operations and accelerate cross-team infrastructure handoffs.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern - (VibrantChem)* *May 2023 – August 2023*

- Automated **Docker** and **Jenkins CI/CD** pipelines for eight microservices on **Azure**, cutting deployment time **50%** while improving observability through **Sentry**, Azure Monitor, and structured logging for production triage.
- Built **REST APIs** and WebSocket sync for 2M+ records on **Azure SQL**, strengthening reliability and performance for high-volume distributed workflows with relational schema design and query tuning.

**LST Capitals** **India**
*Web Developer Intern* *November 2022 – January 2023*

- Built **Flask APIs** and **Databricks** ingestion pipelines for 50K+ records with sub-200ms responses, documenting system behavior and collaborating on integration requirements across 12+ backend endpoints.
- Diagnosed production defects via logging and monitoring, reducing manual handling **70%** while unblocking integration issues independently and supporting cross-platform delivery workflows.

## Skills

**Languages**: Python, Java, SQL, Bash, JavaScript, TypeScript
**Infrastructure**: Kubernetes, Docker, CI/CD, Jenkins, GitHub Actions, AWS, Azure, Linux
**Backend & Data**: FastAPI, Flask, REST APIs, PostgreSQL, ETL pipelines, microservices, Git
**Observability**: Logging, monitoring, Sentry, Azure Monitor, production debugging, incident triage
**Practices**: PyTest, automated testing, Agile, documentation, performance optimization, collaboration

## Projects

**Linguistix** Code | *Python, AWS ECS, Docker, CI/CD*

- Built a containerized **REST API** on **AWS ECS/Fargate** with **Docker** and CodePipeline **CI/CD**, automated validation gates, and production troubleshooting workflows across cloud deployment pipelines.

**FinNexus** Live | *Next.js, TypeScript, Kubernetes, WebSockets*

- Deployed a scalable platform on **Kubernetes** for 50K+ users and 1M+ transactions with real-time updates, secure access controls, and performance-tuned backend services under high transaction volume.

**CPU-scheduling Simulator** Live | *C++, React, Systems Programming*

- Implemented performance-critical **C++** scheduling logic with a **React** interface, handling edge cases and optimizing resource allocation across 1K+ simulated process combinations for systems-level exploration.